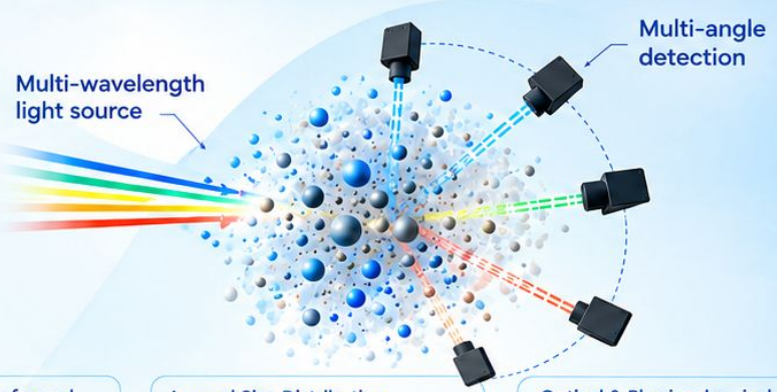
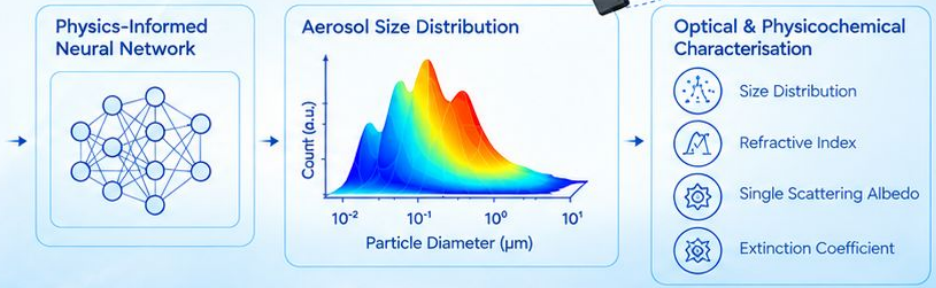
Multi-Wavelength, Multi-Angle Optical Scattering Platform for Aerosol Size-Distribution Retrieval

Physics-Informed Neural Networks + Custom Optical Hardware
for Aerosol Size Distribution & Optical Characterisation

"Air quality monitors tell you how dirty the air is.
We tell you exactly what is in it in real time."





INTELLECTUAL PROPERTY FILED
Patent Pending · Indian Patent Application No. 202621106781
Filing Date: 04 September 2026



MARKET GAP

Air Quality Monitoring Tells Us "How Much" — Not "What"

Today's solutions offer either affordability or detail — not both.



Low-Cost Sensors

₹2,000 – ₹10,000

- Measures estimated PM2.5 / PM10 mass
- Poorly resolves <300 nm particles
- Cannot identify particle material / chemistry
- Sensitive to humidity / dust; requires calibration

Laboratory Instruments

₹50 lakh – ₹1.5 crore+

- Measures detailed particle-size distributions
- Excellent nanoparticle resolution
- Primarily provides size; advanced characterization requires separate techniques
- Slow, lab-bound, expensive; specialized consumables/infrastructure

Our Platform

Compact & accessible

- Measures particle count, size, spread, material & soot
- High-resolution detection from ~10 nm
- Material identification — salts, organics, silica, soot (Limited Capability)
- Real-time, automated, no hazardous consumables

THE MARKET GAP

Today's market forces a trade-off between **affordability, information, and speed.**

Low-cost monitors
Affordable, but limited information

Laboratory instruments
Rich information, but expensive and inaccessible

Our platform
Real-time particle-level intelligence in a compact device

THE OPPORTUNITY

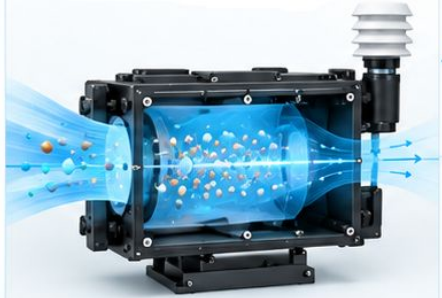
Bring laboratory-grade particle characterization into a real-time, mass-deployable platform — at a fraction of laboratory-instrument cost.

THE SOLUTION

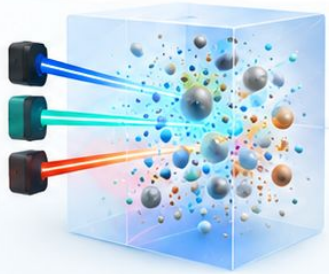
Multi λ - Multi θ Laser Scattering + AI Framework

How Our Device Sees the Invisible

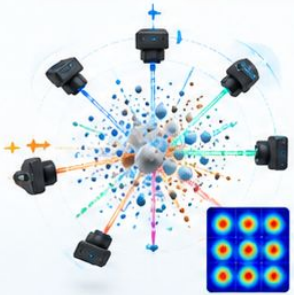
1 Continuous Air Flow



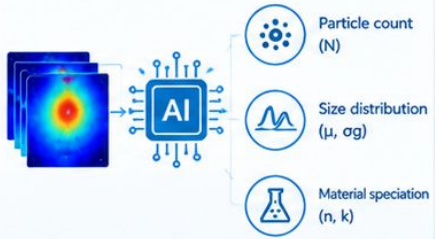
2 Multi Wavelength Laser Probing



3 Multi Angle Light Collection



4 AI Based Decoding



Tested on **1,500,000+** realistic air and aerosol conditions across diverse particle size distributions.

Urban & Industrial | Rural & Remote | Seasonal Variation | Wide Size Range

Model Performance (MAPE)

Target	MAPE
N	0.87 %
μ	0.29 %
σg	0.05 %
n	0.01 %
k	1.86 %

Patent Pending · IN Patent App. No. 202621106781

THE HARDWARE PRODUCT

Compact, Real-Time Diagnostic Instrument

High-performance. Low footprint. Designed for real-world deployment.



*Representative Image

Commercial Hardware Advantages

Compact Desktop or Wall-Mount

Small form factor suitable for cleanroom walls, factory workstations, or hospital rooms.

Real-Time

Updates size and count every few seconds, No waiting 5-10 minutes for slow scans

Zero Recurring Consumables

No toxic chemicals (butanol/alcohol), no radioactive neutralizers, no filters to replace every week.

On-Device Edge AI

Runs entirely inside the device on a low-cost microprocessor. No internet or expensive cloud required.

Disruptive Economics

Built using solid-state lasers and sensors. Enables mass-market pricing at a tiny fraction of lab instruments.

Flexible Deployment





Advanced Air Quality Intelligence in a Compact, Affordable Device

Clean Cities · Healthier Hospitals · Safer Workplaces · Cleaner Environment

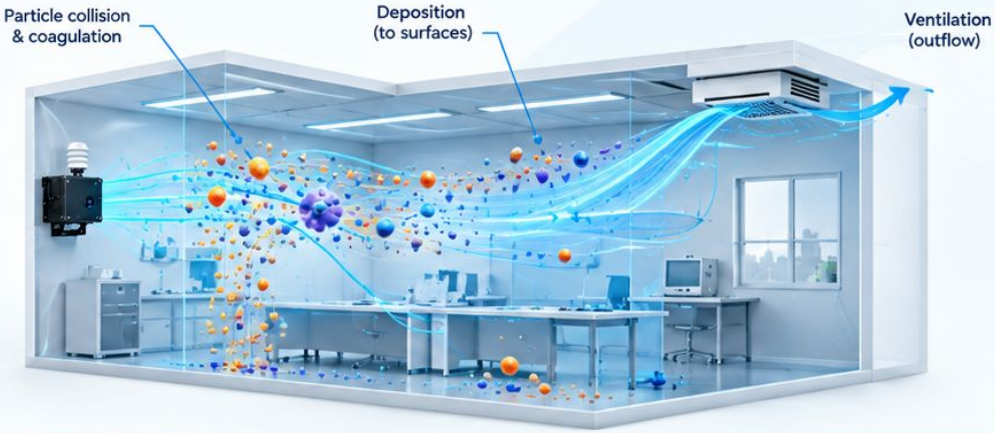
Patent Pending · IN Patent App. No. 202621106781

OUR SECRET SUPERPOWER

Aerosol Dynamics: Tracking How Air Changes Over Time

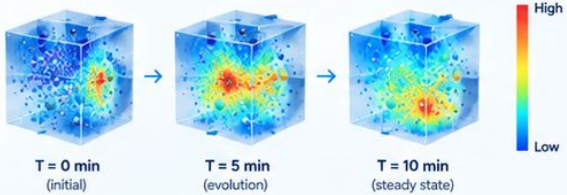
Air Changes Every Minute. We Track Its Evolution.

Airborne particles don't stay still. They collide, merge into bigger clusters, stick to walls, and blow out through air vents.



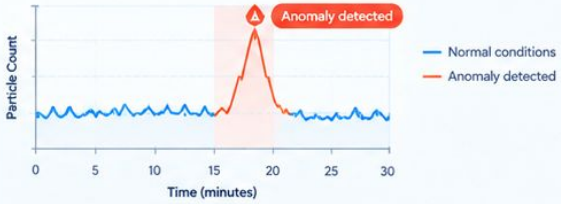
The "Air Digital Twin"

Our system constantly monitors the room's particle evolution to understand natural airflow, deposition, and room ventilation.



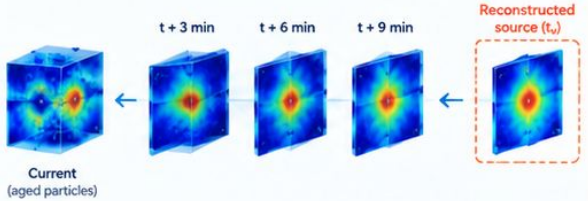
Instant Contamination Detection

Because the AI understands normal room physics, any sudden change (a chemical leak, torn filter, or smoke puff) is flagged instantly.



Time-Reversal Source Tracking

By analyzing aged particles over time, the system can "rewind the clock" to reconstruct the original concentration and exact minute the leak occurred.



THE POWER OF PHYSICS + AI

Not just what's in the air — but how it got there.

Current scattering data + future time steps
+
Aerosol dynamics physics
→
Original aerosol distribution

Patent Pending · IN Patent App. No. 202621106781

REAL-WORLD USES

High-Stakes Applications: Where Failure is Not an Option

Where Customers Need This Urgently





Residential & Indoor Settings

- ✓ Monitor indoor air quality
- ✓ Detect cooking emissions



Occupational & Workplace Settings

- ✓ Monitor worker exposure
- ✓ Detect welding fumes and dust



Ambient Air Monitoring & Hotspot Detection

- ✓ Map air quality in real time
- ✓ Identify pollution hotspots



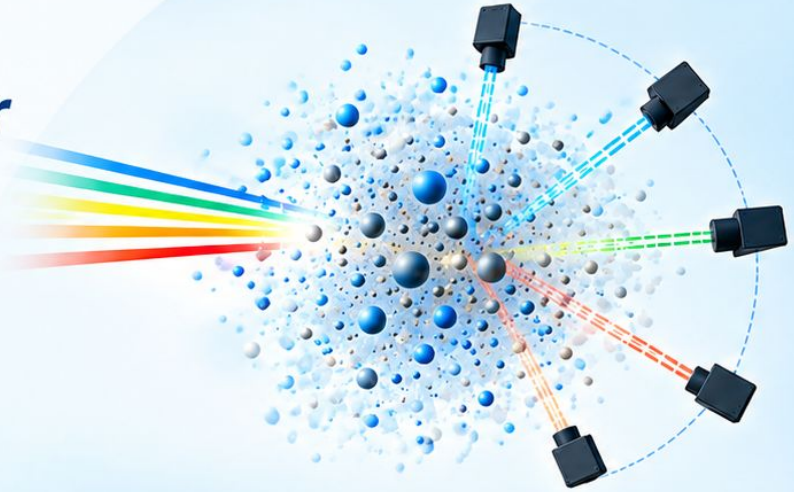
Industrial Factories & Worker Safety

- ✓ Real-time alerts for toxic dust
- ✓ Ensure safer workplaces

Cleaner Air. Safer People. Healthier Industries.

- Protects High-Value Assets
- Ensures Regulatory Compliance
- Safeguards Human Health
- Enables a Cleaner Future

“Using aerosol science to better understand the tiny particles polluting our air.”



Thank You

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